



SUSTAINABLE OLYMPIAD

Sustainable Olympiad 2025

Round 1: Online Knowledge Challenge

Senior Division (ages 15-18)

Mark scheme and worked solutions

Total: 60 marks. Accept answers within rounding and correct alternative methods. Apply error carried forward (ECF): if a wrong value from an earlier part is used correctly in a later part, award the method marks.

Part A: Answer key

Question	Answer	Correct option
A1	B	1.44 kWh
A2	B	Methane has an atmospheric lifetime of only about 12 years, so most of its warming happens in the first decades
A3	C	16/27
A4	C	3.375
A5	C	26%
A6	C	$50 \text{ kJ m}^{-2} \text{ yr}^{-1}$
A7	B	Aerobic decomposition of dead algae by bacteria
A8	D	Increased emission of infrared radiation from a warmer surface
A9	B	33%
A10	C	10 kW

Part A: Worked solutions

A1. Answer B

Power at peak irradiance: $P = 1.6 \text{ m}^2 \times 0.20 \times 1000 \text{ W m}^{-2} = 320 \text{ W}$. Energy: $320 \text{ W} \times 4.5 \text{ h} = 1440 \text{ Wh} = \mathbf{1.44 \text{ kWh}}$. Option A forgets the 4.5 hours.

A2. Answer B

GWP compares the warming caused by a pulse of gas with the same mass of CO_2 over a chosen time horizon. Methane is removed (mainly by OH radicals) with a lifetime of about 12 years. Averaged over 100 years, its short, intense effect is diluted, while long-lived CO_2 keeps warming. Option C is false: the CO_2 produced has a GWP of 1 by definition.

A3. Answer C

Extracting all the energy would stop the air behind the rotor, so no more air could flow through. Momentum theory gives an optimum when the wind is slowed to one third of its upstream speed, giving a maximum power coefficient of $\mathbf{16/27 \approx 59.3\%}$.

A4. Answer C

$P \propto v^3$, so the factor is $(9.0/6.0)^3 = 1.5^3 = \mathbf{3.375}$. This is why turbines are placed on hilltops and offshore.

A5. Answer C

$\text{pH} = -\log_{10}[\text{H}^+]$, so $[\text{H}^+]_{\text{new}}/[\text{H}^+]_{\text{old}} = 10^{8.2 - 8.1} = 10^{0.1} \approx 1.26$, an increase of about **26%**.

A6. Answer C

Producers 50 000 → primary consumers 5000 → secondary consumers 500 → tertiary consumers **50 kJ m⁻² yr⁻¹**. Tertiary consumers are three transfers away from the producers.

A7. Answer B

When the bloom dies, the huge mass of organic matter is broken down by aerobic bacteria, whose respiration consumes dissolved oxygen faster than it can be replaced. During the day algae photosynthesise and release oxygen, and cooler water holds more oxygen, so A and D are wrong.

A8. Answer D

By the Stefan-Boltzmann law, a warmer surface emits more radiation ($\propto T^4$), which opposes further warming. This is the Planck response. The other three amplify an initial warming, so they are positive feedbacks.

A9. Answer B

Temperatures must be in kelvin: $\eta = 1 - T_C/T_H = 1 - 303.15/453.15 = 0.331 \approx \mathbf{33\%}$. Options A and D come from wrongly using degrees Celsius.

A10. Answer C

Electrical input = 14 kW / 3.5 = 4.0 kW. By conservation of energy, heat delivered = heat extracted + work input, so heat extracted = 14 – 4.0 = **10 kW**.

Part B: Worked solutions and mark scheme

B1. Decarbonising school heating (10 marks)

(a) [2 marks] Gas burned = $108\,000 / 0.90 = 120\,000$ kWh. Emissions = $120\,000 \times 0.20 = 24\,000$ kg = **24 t**.

Marking: 1 mark for dividing by the efficiency; 1 mark for the answer.

(b) [2 marks] Electricity = $108\,000 / 3.0 = 36\,000$ kWh. Emissions = $36\,000 \times 0.25 = 9000$ kg = **9.0 t**.

Marking: 1 mark for the electricity used; 1 mark for the answer.

(c) [2 marks] $(24 - 9.0) / 24 \times 100 = \mathbf{62.5\%}$.

Marking: Allow error carried forward from (a) and (b).

(d) [2 marks] Break-even when $36\,000 \times f = 24\,000$, so $f = \mathbf{0.67 \text{ kg CO}_2 \text{ per kWh}}$. This is roughly the factor of a grid that runs mostly on coal.

Marking: 1 mark for the equation; 1 mark for the answer.

(e) [2 marks] Any two of: the COP drops in very cold weather, exactly when heat demand is highest; the grid's emission factor varies by hour and season; leaking refrigerant is a powerful greenhouse gas; a poorly insulated building may need higher flow temperatures, lowering the COP; extra demand may be met by fossil-fuel plants (marginal emissions).

Marking: 1 mark per valid, explained reason.

B2. Fuels and combustion (8 marks)

(a) [2 marks] $\mathbf{2C_8H_{18} + 25O_2 \rightarrow 16CO_2 + 18H_2O}$ (or $C_8H_{18} + 12.5O_2 \rightarrow 8CO_2 + 9H_2O$).

Marking: 1 mark for correct products; 1 mark for balancing.

(b) [2 marks] $M(C_8H_{18}) = 8 \times 12.01 + 18 \times 1.008 = 114.22 \text{ g mol}^{-1}$, so 1000 g is 8.755 mol. This gives $8 \times 8.755 = 70.04 \text{ mol CO}_2$, and $70.04 \times 44.01 = 3082 \text{ g} \approx \mathbf{3.08 \text{ kg}}$.

Marking: 1 mark for the moles of octane; 1 mark for the mass of CO_2 .

(c) [2 marks] Fuel per km = $0.060 \text{ L} \times 0.74 \text{ kg L}^{-1} = 0.0444 \text{ kg}$. $CO_2 = 0.0444 \times 3.08 = 0.137 \text{ kg} \approx \mathbf{137 \text{ g km}^{-1}}$.

Marking: Allow error carried forward from (b).

(d) [2 marks] Ethanol: $M = 46.07 \text{ g mol}^{-1}$. 1 kg is 21.71 mol, giving $43.41 \text{ mol CO}_2 = 1.91 \text{ kg}$. Per MJ: octane $3.08/44.4 = \mathbf{69 \text{ g MJ}^{-1}}$; ethanol $1.91/26.8 = \mathbf{71 \text{ g MJ}^{-1}}$. Ethanol releases slightly more CO_2 per unit of energy. However, its carbon was absorbed from the air by the crop only months earlier (biogenic carbon), so it is largely balanced by photosynthesis. Net savings depend on fertiliser, farm energy and land-use change.

Marking: 1 mark for both values; 1 mark for the biogenic-carbon explanation.

B3. Estimating and modelling a population (8 marks)

(a) [2 marks] $N = MC/R = 60 \times 75 / 15 = \mathbf{300 \text{ beetles}}$.

Marking: 1 mark for the method; 1 mark for the answer.

(b) [2 marks] Any two of: the population is closed (no births, deaths or migration between samples); marks are not lost and do not affect behaviour or survival; marked individuals mix randomly with the population; every individual is equally likely to be caught.

Marking: 1 mark each.

(c) [1 mark] $N = 61 \times 76 / 16 - 1 = 289.75 - 1 = 288.75 \approx \mathbf{289 \text{ beetles}}$.

Marking: 1 mark.

(d) [3 marks] At $N = 300$: $0.40 \times 300 \times (1 - 300/500) = \mathbf{48 \text{ beetles per year}}$. The rate is a quadratic in N with its maximum at $N = K/2 = \mathbf{250}$, where $dN/dt = 0.40 \times 250 \times 0.5 = \mathbf{50 \text{ beetles per year}}$.

Marking: 1 mark for 48; 1 mark for $K/2 = 250$; 1 mark for 50.

B4. Rainwater harvesting (7 marks)

(a) [2 marks] $450 \text{ m}^2 \times 0.750 \text{ m} \times 0.85 = \mathbf{287 \text{ m}^3}$ (286.9 m^3).

Marking: 1 mark for converting mm to m; 1 mark for the answer.

(b) [2 marks] Demand = $600 \times 2 \times 6.0 \text{ L} \times 190 = 1\,368\,000 \text{ L} = 1368 \text{ m}^3$. Fraction = $286.9 / 1368 = \mathbf{21\%}$.

Marking: 1 mark for the demand; 1 mark for the percentage.

(c) [3 marks] Mass = $286\,875 \text{ kg}$. $E_p = mgh = 286\,875 \times 9.81 \times 12 = 3.38 \times 10^7 \text{ J}$. Electrical energy = $3.38 \times 10^7 / 0.50 = 6.75 \times 10^7 \text{ J} = \mathbf{18.8 \text{ kWh}}$, which is very small compared with the water saved.

Marking: 1 mark for mgh ; 1 mark for the efficiency; 1 mark for the kWh conversion.

B5. Carbon dioxide and global temperature (7 marks)

(a) [2 marks] $\Delta F = 5.35 \ln(420/280) = 5.35 \times 0.405 = \mathbf{2.17 \text{ W m}^{-2}}$. $\Delta T = 0.80 \times 2.17 = \mathbf{1.7 \text{ K}}$.

Marking: 1 mark each.

(b) [1 mark] $\Delta F = 5.35 \ln 2 = 3.71 \text{ W m}^{-2}$, so $\Delta T = \mathbf{3.0 \text{ K}}$.

Marking: 1 mark.

(c) [2 marks] $\Delta F = 1.5 / 0.80 = 1.875 \text{ W m}^{-2}$, so $C = 280 \times e^{1.875/5.35} = 280 \times 1.42 = \mathbf{398 \text{ ppm}}$, a level the atmosphere passed around 2014.

Marking: 1 mark for ΔF ; 1 mark for C .

(d) [2 marks] The climate is not yet at equilibrium: the oceans' large heat capacity delays warming by decades. Aerosol pollution reflects sunlight and masks part of the greenhouse warming. (Other greenhouse gases add warming, so they do not explain the gap.)

Marking: 1 mark per valid reason.